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Proposal for Provider-Agnostic Wallet Base Architecture (RFP-T-BACK-0006)

Submitted by: Systango Technologies Limited

1. Executive Summary

1.1 Introduction

Systango Technologies Limited (Systango) is pleased to submit this proposal in response to Polity's Request for Proposal for the Provider-Agnostic Wallet Base Architecture (RFP-T-BACK-0006, V.2.3.1). We understand that this engagement forms the foundational trust, evidence, and control layer of the R1.3.2 wallet infrastructure stack.

Our primary objective is to deliver a robust, provider-neutral foundation that isolates provider-specific logic, secures credentials behind the Polity Vault REST gateway, enforces strict idempotency, and utilizes Temporal workflow orchestration to make wallet operations deterministic, reviewable, and governable.

Our key architectural and safety commitments include:

- **Zero-Leakage Credential Boundaries:** Eliminating direct secret-store or Infiscal Access from the Wallet Infrastructure Service (WIS), ensuring provider credentials are resolved strictly at Activity execution time via the Polity Vault REST gateway.
- **Fail-Closed Security Gates:** Engineering a multi-tier gate precedence matrix (evaluating global enablement, provider state, environment lanes, and a sensitive-operation kill switch) before any Temporal execution occurs.
- **Strangler Fig Coexistence:** Preserving all current read/display structures for legacy DFNS and Safeheron surfaces while introducing a clean, unified API bridge.
- **Three-Part Sensitive-Operation Proof:** Delivering static code scans, runtime API inventories, and attestation logs to guarantee no unapproved custodial, signing, or transaction-execution paths are reachable.

1.2 Systango: Your Trusted Systems Engineering Partner

Systango is a publicly listed digital engineering company (NSE) with an 18-year history of delivering secure, high-availability enterprise backend architectures. Our **ISO 27001 certification** and verified cloud partnerships across AWS, GCP, and Azure underscore our readiness to manage sensitive transaction layers and high-integrity infrastructure.

We fully accept and embrace Polity's **AI-assisted coding productivity baseline**. Our engineering workflows are optimized for interactive code assistants and agentic environments, managed under strict prompt hygiene and confidentiality guidelines to guarantee that no payload context or code provenance is compromised.

2.Detailed Approach and Methodology:

Our technical delivery treats the codebase as an active enterprise system, avoiding sudden replacements in favor of a strictly controlled migration path.

2.1 Overall Scope & Delivery Framework

The implementation is structured into **six logical review boundaries** matching the mandated MR-BASE namespaces. Each milestone requires verifiable evidence artifacts linked directly to the Work Breakdown Structure (WBS).

Phase / Namespace	Scope & Key Activities	Key Deliverables & Artifacts
MR-BASE-01: Discovery & Freeze	Freeze current-state source; execute complete route, table, and credential inventory within the existing WIS	Route & Table Inventory; Exception Log; Baseline Contract Freeze Note
MR-BASE-02: Contract & Gates	Implement ProviderAdapter and ProviderCredentials interfaces; wire up the AdapterRegistry and multi-tier feature flag kill switches.	Core Go Interface Package; Type-Conformance Compile Proofs; Failure-Gate Unit Tests
MR-BASE-03: Persistence Layer	Execute reversible formigrate scripts for provider-neutral schemas; build the WALLET_OPERATIONS ledger and concurrency controls.	Up/Down Migration Logs; Database Unique Constraint Verification; Rollback Package
MR-BASE-04: Resilient Execution	Bootstrap Temporal Go SDK client/worker boundaries; implement deterministic workflows and activity-time Vault access.	ExecuteWorkflow Integration Tests; Local Worker Smoke Logs; History Secret-Redaction Scan
MR-BASE-05: Unified Routing	Deploy /api/wallets/v1/* unified bridge; implement fail-closed webhook envelope and sensitive-operation constraints.	OpenAPI Route Fate Matrix; Negative Capability Tests; Webhook Verification Logs
MR-BASE-06: Closure & Handoff	Finalize CSV-ready documentation, cross-component verification, and downstream provider integration checklists.	Complete CSV Traceability Index; Operational Runbook; Provider Handoff Pack

2.2 Core Technical Alignment

Our backend engineering directly fulfills the requirements of the BASE-REQ architecture:

- **The Interface Dispatch Boundary:** The AdapterRegistry is implemented as an explicit startup-time check that rejects duplicate provider registrations and converts unknown requests into controlled canonical errors.
- **Temporal & SAGA Topology:** Tier 1 workflows (ProvisionWalletWorkflow and RegisterAccountWorkflow) are separated from Tier 3 Activities. Long-running provisioning chains use forward-recovery and reverse-order compensation logic driven by the temporal-saga-go framework.
- **Fail-Closed Webhook Boundary:** The inbound boundary normalizes payloads to raw bytes to preserve signatures while validating authenticity, replay protection, and sequence timestamps prior to mutating any internal state.

2.3 QA and Evidence Methodology

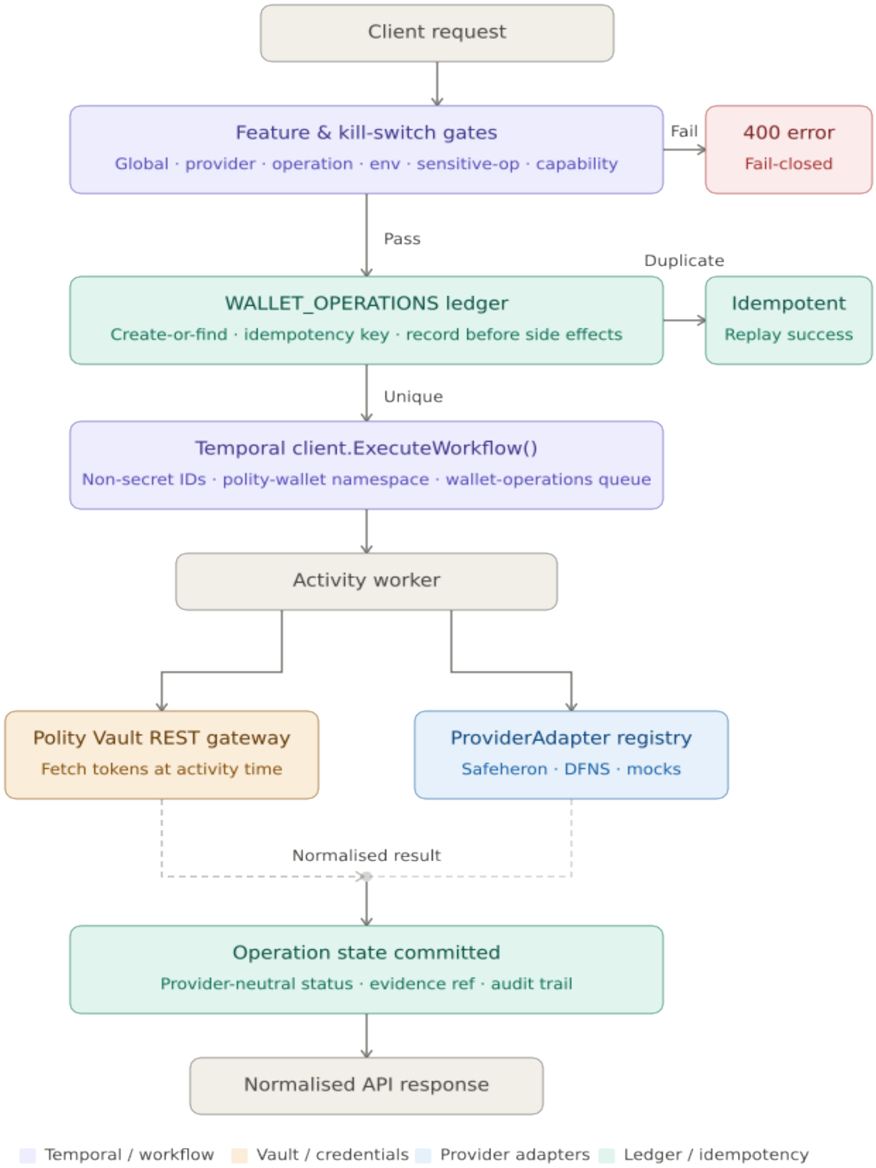
We do not treat quality assurance as an afterthought; a task is only complete when its machine-verifiable evidence is generated.

- **Static Vulnerability & Route Scans:** Regular executions of golangci-lint and automated structural search tools to ensure no unapproved routing parameters or exposed admin methods escape isolation.
- **History Replay Validation:** Programmatic testing of workflow definitions against historical mock logs to prove absolute determinism and verify that zero secret tokens leak into the Temporal execution state.
- **Data Consistency Testing:** Live dry-runs of data schema migrations and structural rolling rollbacks to confirm that historical records remain safe and clean during deployment.

2.4 DevOps & Governance Alignment

- **Version Control:** Git branching per Polity DevOps and development workflow baseline. Branch naming follows MR-BASE-NN convention (e.g. feature/wallet-base/base-01-inventory-contract-freeze).
- **MR Standards:** Single-purpose MR packages; oversized MRs carry written rationale tied to migration atomicity, Temporal boundary integrity, or sensitive-operation closure.
- **Stop Conditions:** Each MR description records the stop condition, any blocked-condition register entries, and variance against the accepted repository state.

- **Traceability:** Full audit trail from BASE-REQ/BASE-FR/BASE-NFR → WBS row → MR-BASE package → test evidence → QA checklist → evidence index, without orphaned entries.
- **AI-Assisted Coding Controls:** We will apply prompt-hygiene controls per the Confidentiality and NDA metadata row. No Polity Confidential Information, secrets, Vault references, workflow history, signing payloads, raw provider keys, or live database extracts will be transmitted to any third-party AI provider. All AI-produced artefacts are reviewed and owned by our engineers.



3. Deliverables and QA Artefacts

Acceptance will be driven by an indexed delivery pack containing the following hard engineering components:

- **Codebase Adaptations:** Modular, Vite-clean/Go-conforming packages within the authoritative WIS repository covering interfaces, registries, and workers.
- **Database Migration Suite:** Verified up/down gormigrate execution logs demonstrating clean schema instantiation and rollback capabilities.
- **Route Fate Matrix:** A comprehensive documentation map listing the status (preserved, wrapped, or blocked) of every production endpoint.
- **Static & Dynamic Sanitization Proofs:** Automated file scans proving that no reference credentials, static tokens, or private signing keys exist in the persistent store, console output, or tracking history.
- **CSV-Ready Traceability Index:** A complete spreadsheet cross-referencing every WBS line item directly to its corresponding requirement token, merge block, and confirmation log.

3.1 Out of scope

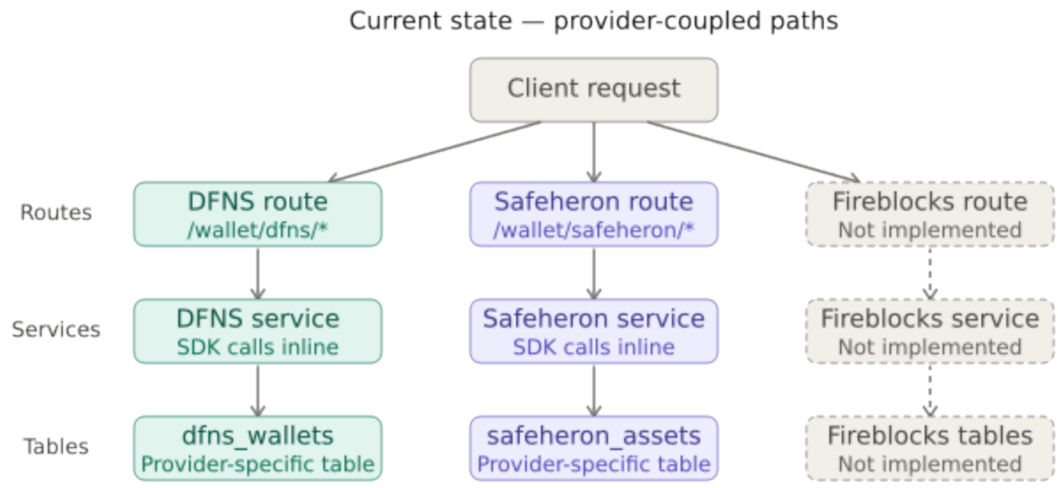
- Provider SDK/runtime activation and production provider accounts
- Production signing, payment, raw broadcast, WalletConnect, or transaction execution
- Temporal Server / EKS / Helm / ArgoCD provisioning (platform scope)
- Nexus activation
- Direct WIS access to Infisical; secrets in workflow inputs, history, logs, or ordinary wallet rows
- Frontend production cutover via the new base service until provider RFPs complete
- New chains or payment rails outside the approved release chain set without governance approval

4. Architecture

4.1 High-Level View

Today, wallet behavior is directly tied to DFNS and Safeheron paths across routing, services, and tables. The target architecture establishes a single provider-neutral operation path while keeping legacy routes alive as compatibility wrappers during the transition phase (**Strangler Fig Migration strategy**).

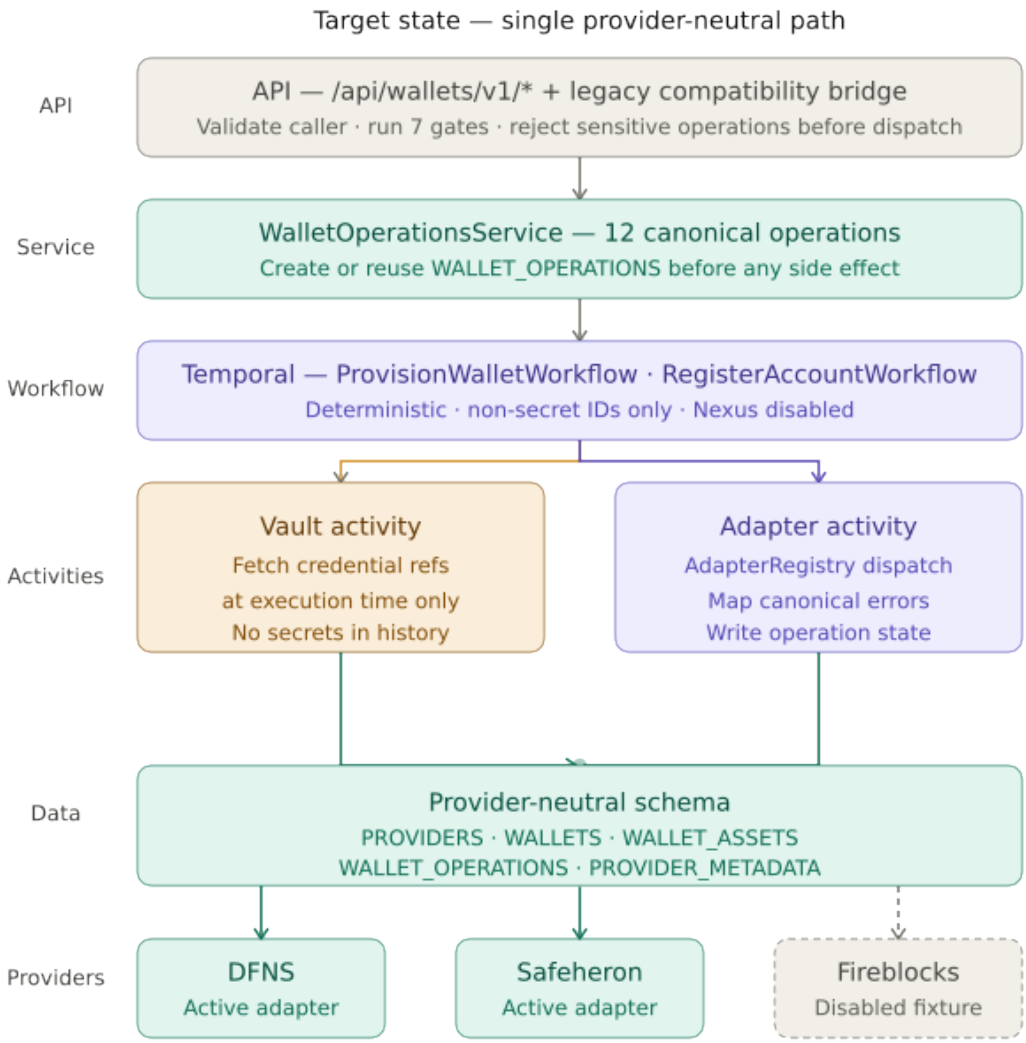
Current State: Fragmented Integration Architecture



Current State Architecture

Target State : Request Flow (Unified Operation Lifecycle)

Every single wallet change or lifecycle event follows an unyielding, deterministic execution pipeline before mutating downstream state



Target State Architecture

4.2 Target Layer Matrix

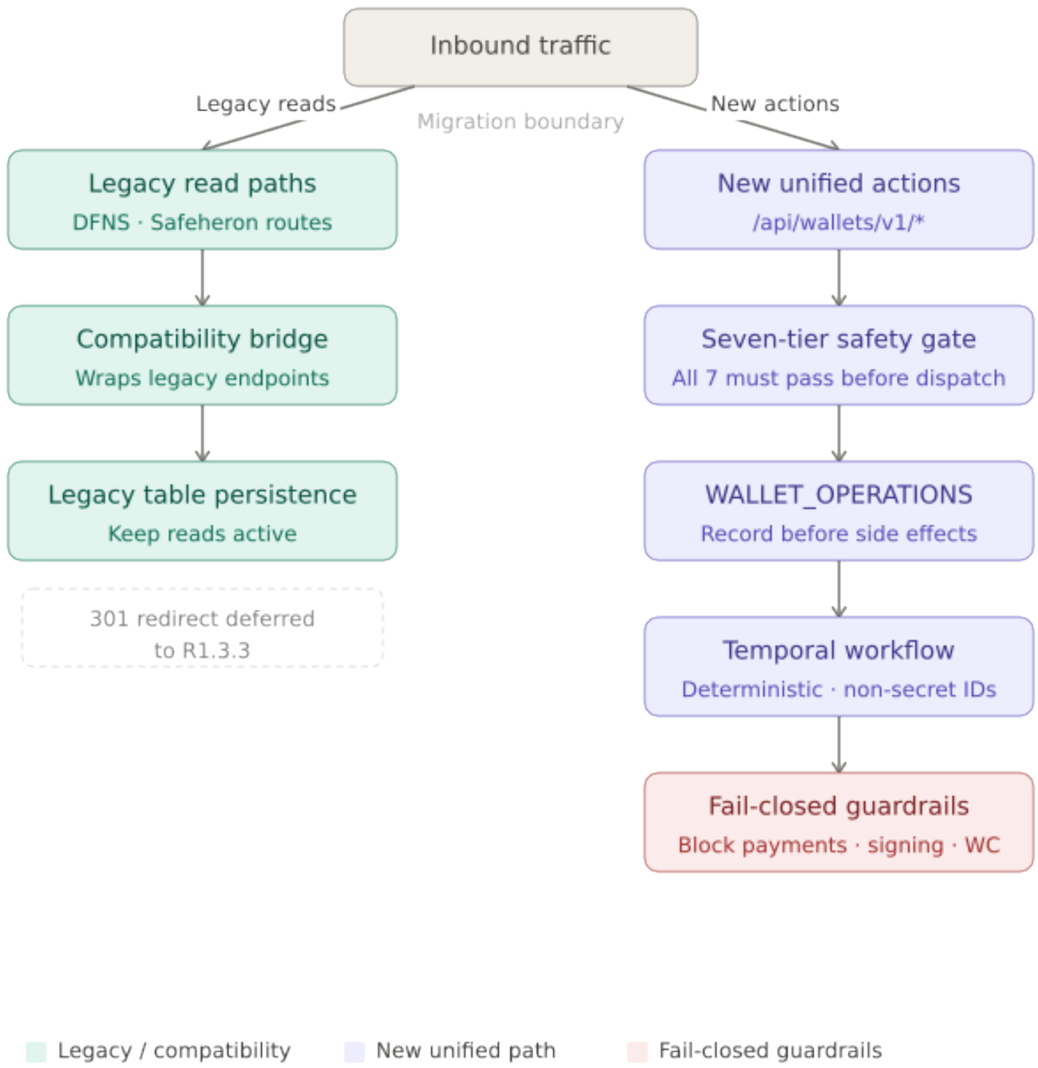
Layer	System Responsibility
API	Validates caller context, runs feature and security gates, and exposes the unified /api/wallets/v1/* surface or bridges to legacy read routes.
Service	Handles structural request validation and creates or reuses WALLET_OPERATIONS ledger entries prior to triggering any external side effects.
Workflow	Orchestrates downstream lifecycle loops through direct Temporal Go SDK execution; carries deterministic, non-secret payloads only.
Activities	Executes all external platform side effects: Vault gateway invocations, provider adapter routing, database updates, and callback handling.
Data	Enforces a robust, provider-neutral database schema combined with a strict operational ledger to prevent replay or race conditions.
Providers (Later)	Downstream provider workstreams (Fireblocks, DFNS, Safeheron RFPs) plug cleanly into the frozen base interface contract without redefining the shared core.

Core Building Blocks

- **WalletOperationsService:** Exposes 12 canonical operations. Account management, wallet structures, and asset lifecycles are fully active in R1.3.2; transaction capabilities are defined at the contract level but explicitly deferred.
- **ProviderAdapter & AdapterRegistry:** Forms a single execution-time dispatch boundary for all provider modules. The registry automatically rejects duplicate provider registrations at boot time and returns failure-safe unknown-provider errors for unmapped targets. Fireblocks is introduced purely as a disabled baseline fixture.
- **The Seven-Tier Safety Gates:** Compels every request to pass a sequence of checks before workflow execution can begin: (1) Global Wallet status, (2) Provider enablement, (3) Operation permission, (4) Environment/segment lane constraints, (5) Sensitive-operation kill switch status, (6) Provider technical capability, and (7) Idempotency/state validation.
- **Polity Vault REST Integration:** Restricts all credential resolution to Activity execution time. Workflows and persistent database fields store tokenized reference references only—never raw secrets, private keys, or API credentials.

4.3 Coexistence & Strangler Fig Transition Mechanics

To keep the production system completely stable while re-architecting the base layers, the execution pipeline enforces strict isolation between legacy models and unified endpoints.



Coexistence & Strangler Fig Transition Mechanics

- **Read Compatibility Guarantee:** Existing DFNS and Safeheron wallet read pathways remain actively supported through an isolated database and routing compatibility bridge to ensure zero client-side breakage.

- **Sensitive Route Elimination:** Prohibited and custodial endpoint families (such as active token payment execution, message signing, and WalletConnect invocation paths) are explicitly removed, deactivated, or configured to fail closed under negative capability testing.
- **Downstream Provider Isolation:** Provider-specific crypto engines, sandbox configurations, and web SDK bridges remain completely deferred to downstream work packages, ensuring zero scope creep during base architecture construction.

5. Tech Stack:

Layer	Technology	Role in this engagement
Language	Go 1.24	Matches current wallet-infrastructure-service
HTTP API	fasthttp, fasthttp/router	Existing WIS entrypoints; new unified routes added alongside legacy
Database	PostgreSQL, GORM, gormigrate v2	New provider-neutral migrations; reversible up/down
Cache	Redis (go-redis)	Existing use; credential handling moves to reference-only at Activity boundary
Orchestration	Temporal Go SDK (new)	Workflows, activities, worker; local Docker Compose for proof
Secrets	Polity Vault REST (providers/polityVault)	Activity-time credential access; no direct Infisical from WIS
Feature flags	Viper (flagger/)	Extended into full gate precedence matrix
Auth (compatibility)	JWT/RBAC in WIS; Dynamic v4 on frontend	Non-regression samples/tests only; no frontend redesign
Quality	go test, golangci-lint, static scans	Per WBS-BASE-26 and MR evidence packs
Integration	polity_tests (where in scope)	Supporting evidence for release boundary
New code areas	internal/contract, cmd/worker/main.go, provider-neutral migrations	Per RFP Appendix A contract shapes

6. Project Plan and Timeline

Our planning is explicitly calibrated against the 90 planned focused engineering hours baseline under the assumption of AI-assisted engineering productivity.

Milestone	Target Timeline	Deliverables / Activities
Project Kick-off	T	Repository access validation, environment initialization, and baseline sync.
Discovery & Contract Freeze	T + 4 Days	Delivery of MR-BASE-01, concluding WBS items 01–03 with an approved route inventory.
Mid-Term Functional Delivery	T + 12 Days	Integration of core contracts, registries, database migrations, and WALLET_OPERATIONS persistence (MR-BASE-02 and MR-BASE-03).
Final QA Submission	T + 20 Days	Successful execution of Temporal workflows, worker hooks, unified API bridge, and all negative capability verification vectors (MR-BASE-04 and MR-BASE-05).
Final Acceptance & Sign-off	T + 25 Days	Processing of the final technical review, delivery of the CSV index, runbooks, and handover signature (MR-BASE-06).

7. Project Team and Expertise

Every assigned member of our core squad brings extensive experience managing high-integrity backend data systems and distributed system state.

Role	Experience	Responsibilities
Lead Backend & Distributed Systems Developer	10+ Years	Responsible for building the Go interface contracts, wiring up client-side Temporal execution loops, and configuring core SAGA orchestration mechanics.
QA & System Validation Engineer	10+ Years	Responsible for verifying strict idempotency constraints, generating negative capability proofs, organizing endpoint inventories, and running history replay tests.

8. Commercial Proposal

8.1 Fixed-Price Summary

Description	Estimated Hours	Blended Rate (USD/hr)	Total (USD)
Total Development & QA Effort	90	38	3,420

Total Fixed Price (USD): $3,420

(Inclusive of all vendor overheads, profit, and administrative costs.)

8.2 Payment Milestones

Milestone	Percentage	Condition
Kick-off	20%	Contract signing and environment setup
Mid-Term Delivery	20%	50% of WBS is complete with design validation
Final QA Submission	30%	All items implemented and QA approved
Final Acceptance	30%	Final sign-off by Polity Design & Engineering Teams

9. Assumptions, Dependencies, and Blocked Conditions

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9.1 Assumptions

- Polity provides repository access, the engagement pack (TOR AC/Gate definitions, approved release chain set), and written answers to clarifications within the RFP window.
- Polity Vault REST is available for integration testing, or Polity agrees to signed-off mocks for Vault and provider fixtures.
- Local Temporal (Docker Compose or equivalent) is available for MR-BASE-04 proof.
- DFNS and Safeheron mock adapters are the primary acceptance path; live sandbox evidence is additive.
- Delivery is remote; AI-assisted coding is used with the developer reviewing all generated code.
- Work outside WBS-BASE-01..26 requires written change-control approval.

- All estimates and implementation commitments are based on the assumptions defined in the [Feature list with estimated hours and considerations]

9.2 Dependencies

Dependency	Provided by	If unavailable
Engagement pack and TOR gate definitions	Polity	Evidence scope unclear; blocked register until resolved
Polity Vault REST or approved mock	Polity / environment	BASE-08 and BASE-17 blocked or mock-only
Local Temporal runtime	Vendor setup / Polity docker-compose	BASE-14..18: local proof + blocked note
Go 1.24 toolchain and golangci-lint in CI	Environment	BASE-26 incomplete
Dynamic / FE–Auth coordination for BASE-22	Polity	Compatibility evidence delayed

10. Optional Support and Maintenance

Systango Technologies Limited (Systango) can provide post-implementation technical support for one (1) month after final acceptance. This includes: - Minor UI/UX fixes - Bug resolution for R1.3.2 features - Handover support to Polity's R1.3.3 Circle Team

Item	Duration	Cost (USD)	Description
Post-Go-Live Support	1 month	$3040	Bug fixes and QA support

11. Proposal Acceptance

We thank Polity for this opportunity and affirm our commitment to delivering a secure, performant, and visually consistent frontend that aligns with the R1.3.2 objectives and standards.

Authorised Signatory:

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